Data Analysis & ETL Report: Final Summary

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Short description

In this document we analyze a dataset constituted of three different csv files:

- **Account-statement-1-1-2024-12-31-2024.csv**, containing the type of transaction done by a firm at an instant of time.
- **country.csv**, containing the geographic location of the firm and all the geographic details.
- **symbols.csv**, containing the main characteristics of the firm.

Our analysis focuses on trying to create a model for the data warehouse and then analyze the data to answer several analytical questions.

1 Data Warehouse Modeling: Star Schema

In this section we analyze the different aspects of this dataset. First of all we establish the fact grain, then we create a tree schema to better understand the relations between tables, furthermore we identify the fact and other dimensions (in this step we didn't delete any features of tables) and at the end we created the star schema.

2 Data Transformation and Analysis

In this part we clean the data and then answer various analytical business questions. In the first part we load the three different files and do a first analysis. Apparently, the transaction dataframe has 464 missing values and 2745 missing values for an "Unnamed" feature. This is due to a mistake during the CSV creation: first of all, at the end of the file, there are 464 ";;;;;" rows that Pandas recognizes as actual rows. Furthermore, at the end of each valid line, there is an extra ";", which explains why another column named "Unnamed" was created.

For our analysis, we do not need the subregion or the intermediate region, so the presence of missing values for them doesn't matter. We can see that for the 'alpha-2' code, only Namibia has missing values; therefore, we will use 'alpha-3' for our analysis instead. The region 'Antarctica' can be removed from the dataset. Additionally, Taiwan lacks both a 'region-code' and a 'region'. This is likely due to geopolitical and geo-economic reasons; however, since we know it is located in Asia, we will manually update these missing fields to ensure our data is complete.

Then we notice 212 firms that do not have a correspondent in the symbols dataframe. two distinct approaches could be pursued at this point:

- Give as value to everything "Unknow".
- Delete the rows containing that firm.

I decided to choose the second option. The purpose of this choice is to not affect the analysis and prevent from biasing the analysis.

In the step 2.2 we answer to 5 different questions:

1. **What are the top 5 sectors by number of SELL transactions in US during 2024?** Technology dominates the market of sells in 2024.
2. **What are the top 10 countries by number of SELL transactions in 2024?** USA dominates the market of sells in 2024.
3. **What are the top 5 sectors by total units traded (BUY + SELL) in Q3 of 2024?** In Q3 of 2024 technology is best trader.
4. **What are the top 5 countries by number of units traded in Financial Services companies?** USA is the best Financial Services trader in 2024.
5. **What are the top 5 industries by units traded in Asia in 2024?** In Asia semiconductor is the best trader in 2024.

The answers are are provided printing the final dataframe and displaying a bar chart.

3 Streamlit Dashboard

In this section we create a dashboard using streamlit to have a better interaction and perform a deeper analysis using different period of time.

4 Conclusion

The Homework 3 goal is to simulate the responsibility of a data analyst: definining the ETL process, creating a useful dataframe, analyzing the data to answer different business questions and building a useful dashboard. All these objectives were successfully achieved and the insights and answers visualized through distinct bar charts.